CS23E01 / EMBEDDED SYSTEMS AND INTERNET OF THINGS

Session: July-Nov 2026 (B.E. CSE – VII sem)

Exercise for Lab-2:

Exercise 1:

Create a program that repeatedly checks a control variable.

Choice = 1 → Addition
Choice = 2 → Subtraction
Choice = 3 → Multiplication
Choice = 4 → Division
Choice = 5 → Exit

Each operation should be implemented as a subroutine.

Exercise 2:

Count Number of 1's in an 8-bit Number.

Exercise 3:

Display a counter that counts down from 10 to 0 with 1 second delay.

Exercise 4:

Perform 16-bit Addition/Subtraction of Two Numbers.

Exercise 5:

Control LED and motor using bit-addressable RAM.

Exercise 6:

Reverse an Array Using Stack (PUSH/POP).

Exercise 7:

Switch between register banks using PSW.

Exercise 8:

Read and Display P0, P1, ACC, B, PSW, SP, DPTR.